Katharina Alicia Krenzer

M.Sc. Statistics · ETH Zürich

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LinkedIn · GitHub

My research starts from a gap between the clinic and everyday life. A hospital record sees patients only when they seek care, while wearables record the months in between. I build mixed-initiative AI that knows when to ask for human expertise and shows its uncertainty so clinicians can question it, drawing on projects with ENT, intensive care and spinal cord injury clinicians.

01 PUBLICATIONS

NeurIPS 2026
Workshop · Accepted

GNN-VIEW: Visual Inspection and Explanation of Graph-based Earth-System Models

A. Lucatelli, F. Unterhuber, **K. Krenzer**, M. El-Assady, B. Soja, C. Humer, J. Oskarsson

Workshop on Advances in Representation Learning for Earth Observation. Visual analytics for inspecting and explaining graph-based earth-system models.

IEEE VIS 2026
vis4climate · Accepted

GNN-VIEW: Visual Inspection and Explanation of GNN-based Weather Models

A. Lucatelli, F. Unterhuber, **K. Krenzer**, L. Kummer, J. Oskarsson, C. Humer

Interactive visualisations and experiments that make the behaviour of GNN weather models inspectable, with SMHI and ECMWF.

02 EDUCATION

2025 – 2027 (exp.)
Zurich, CH

M.Sc. Statistics

ETH Zürich

GPA 5.5 / 6.0. Focus: machine learning for healthcare, interactive visualisation and clinical decision support.

2022 – 2025
Würzburg, DE

B.Sc. Economics & Computer Science

University of Würzburg

Grade 1.2 (best 1.0). Graduated best in cohort in five semesters.

Exchange
Bergen, NO

Exchange Semester

NHH Norwegian School of Economics

Grade A.

03 RESEARCH EXPERIENCE

09/2026 – present

Research Assistant & Project Lead: Nasal Polyp Active Learning

ivia Lab, ETH Zürich with University Hospital Zurich

Developing the detection model and active-learning pipeline, including acquisition functions and evaluation, with ENT clinicians. The model flags uncertain endoscopy frames for clinician review.

09/2026 – present

Spinal Cord Injury Decision Support

BMDS Lab, ETH Zürich with Schulthess Klinik

Developing and evaluating a clinical dashboard that combines CNN recovery predictions and patient clustering to compare a patient's recovery with similar patients.

02/2026 – present

Research Collaboration: GNN-VIEW

ivia Lab, ETH Zürich with SMHI and ECMWF

Implemented GNN visualisations and experiments; prepared ECMWF training materials.

03/2026 – 09/2026

Research Project: Fairness in ICU Foundation Models

Biomedical Informatics Group, ETH Zürich

Analysed demographic fairness in the ICU foundation model ICareFM and showed how differences in event prevalence drive apparent group disparities.

2023 – 2025

Research Assistant: Information Systems & Business Analytics

University of Würzburg

Applied graph neural networks to spatiotemporal logistics data and studied generalisation under distribution shift.

04 INDUSTRY & TEACHING

04/2025 – 09/2025
Ingolstadt, DE

Data Science Intern: AI & Data-Driven Development

AUDI AG

Built ML pipelines and FastAPI services, including RAG-based classification and workflow automation.

04/2024 – 09/2024
Würzburg, DE

Teaching Assistant: Econometrics

University of Würzburg

Led statistics and econometrics tutorials for approximately 50 students.

05 HONORS & SERVICE

2025 – 2026

e-fellows Scholarship

2023 – 2025

Dean's List

University of Würzburg · Top 3%

ETH Zürich

Elected Statistics Student Representative

One of six representatives. Organises cohort trips and events for women in the programme; designed and produced the cohort merchandise.

ETH Zürich

Google Developer Group on Campus · UN Student Team

UN Student Team project on conflict network analysis.

06 SKILLS & LANGUAGES

ML & Statistics: Python · PyTorch · PyTorch Geometric · TensorFlow · scikit-learn · R · SQL

Engineering: FastAPI · Git · Docker · Azure · LaTeX

Languages: **German** Native · **English** C1 · **French** DELF